BEEE IAT2 QB (2025-26) AIML A and B

3M Questions

1. Explain the working Principle of single phase transformer with neat sketch.
2. Derive the EMF equation and hence transformation ratio for the single phase transformer.
3. Define the efficiency of transformer and list any four applications of a transformer.
4. Explain the various transformer losses.
5. Compare between Core type and Shell type transformers. (Any three)
6. State the efficiency of a transformer. List the applications of transformer.

7M Questions

1. State the relationship between phase and line voltage as well as current for star and delta connected balanced load across a three-phase balanced system. A balanced three phase star connected load consist of three coils, each consisting of R=6 ohms and XL = 8 ohms. Determine line current when the load is connected across 400 volts, 50 Hertz's supply.
2. An inductive coil having inductance of 0.04H and resistance 25 ohms has been connected in series with another inductive coil of 0.2H and resistance 15 ohms. The whole circuit is powered with 230 v, 50 Hz mains. Calculate the power dissipation in each coil and total power factor
3. With neat sketch explain the working principle of DC motor.
4. Derive the formula for resonant frequency of a series resonant circuit. Compare series and parallel resonance
5. Two impedances of 12 + j16 ohm and 10 - j20 are connected in parallel across 230V, 50Hz, single phase ac supply. Find admittance, current and power factor of each branch.
6. State the relationship between phase and line voltage and current for delta connected balanced load across a three-phase balanced system. A balanced three phase delta connected load with per phase impedance of 20 ohms. Determine line current and phase current ,VL=100 V.
7. Explain the principle of operation along with constructional details, three phase induction motor. List its types and applications.
8. Define the following terms for single phase AC supply with suitable waveform. (i)RMS and Average value (ii) Amplitude (iii) Cycle (iv) Time period (v) Frequency.